

Lecture #23

Content to be covered

- ❖ Wave Packets
- ❖ Wave velocity and Group Velocity
- ❖ Relation between phase velocity and group velocity

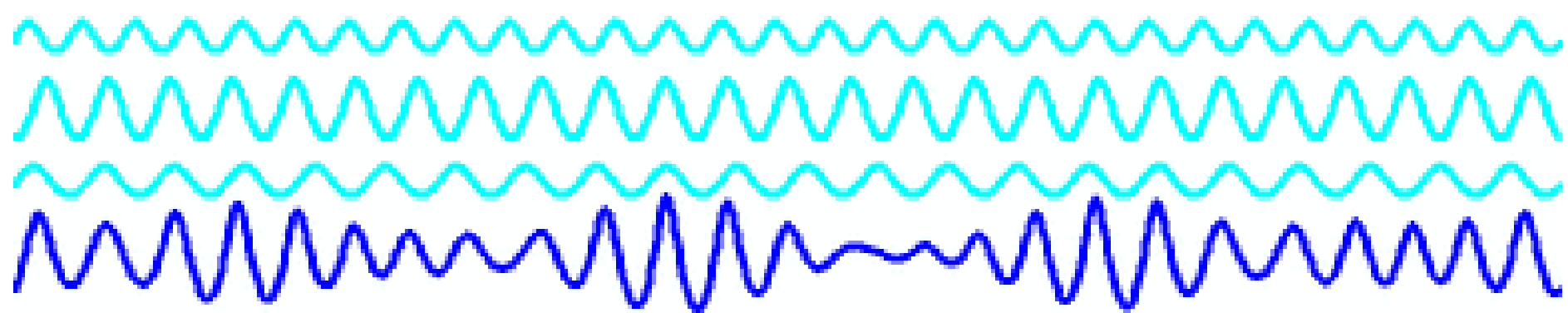
Resources to be consulted

- ❑ Concepts of Modern Physics by Arthur Beiser

Addition of Two Waves

- Real waves are almost always combinations of multiple wavelengths
- Average these two expressions to get a new wave:

$$Y_1 = A \cos(k_1 x - \omega_1 t) \text{ and } Y_2 = A \cos(k_2 x - \omega_2 t)$$



- This wave has two kinds of oscillations:

The oscillations at **small scales** and The “lumps” at **large scales**

$$Y = A \cos(\bar{k}x - \bar{\omega}t) \cos(\Delta kx - \Delta \omega t)$$

**Small scale
oscillations**

**Large scale
oscillations**

$\Delta k = \frac{1}{2}(k_1 - k_2)$

$\Delta \omega = \frac{1}{2}(\omega_1 - \omega_2)$

$\bar{k} = \frac{1}{2}(k_1 + k_2)$

$\bar{\omega} = \frac{1}{2}(\omega_1 + \omega_2)$

Group Velocity

$$Y = A \cos(\bar{k}x - \bar{\omega}t) \cos(\Delta kx - \Delta \omega t)$$

Small scale
oscillations

Large scale
oscillations

The velocity of little oscillations governed by the first factor

• Leads to the same formula as before for phase velocity:

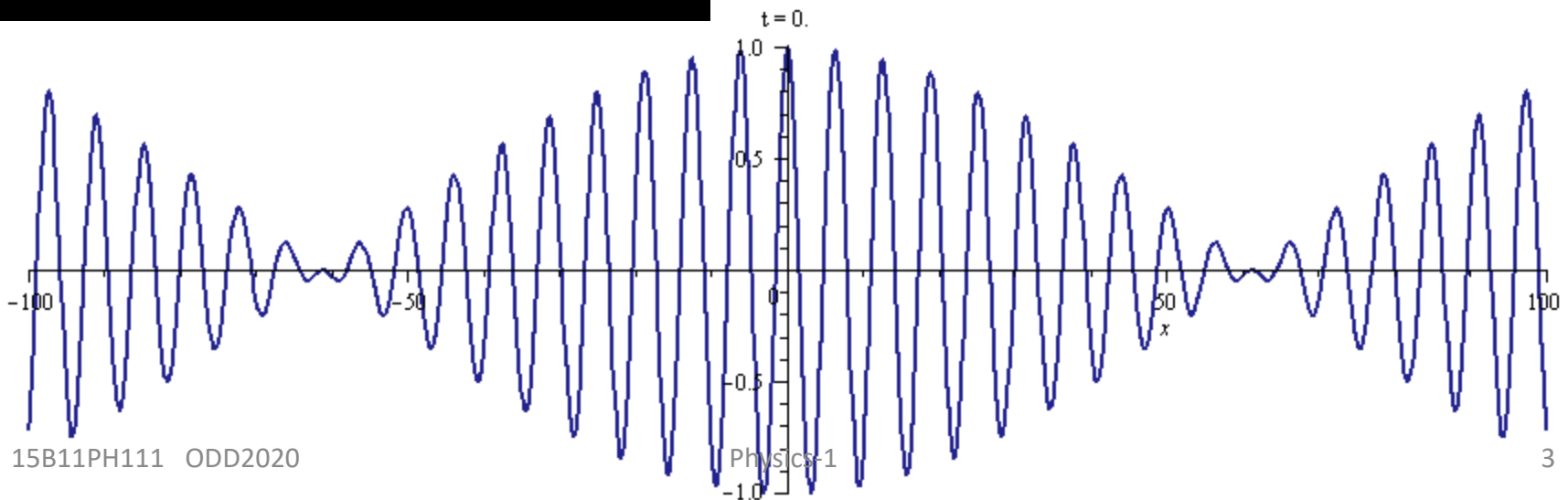
$$v_p = \frac{\bar{\omega}}{\bar{k}}$$

The velocity of big oscillations governed by the second factor

• Leads to a formula for group velocity:

$$v_g = \frac{\Delta \omega}{\Delta k}$$

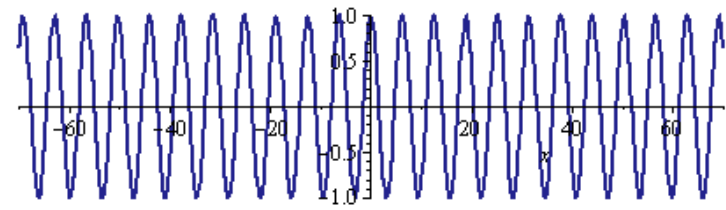
These need not be the same!



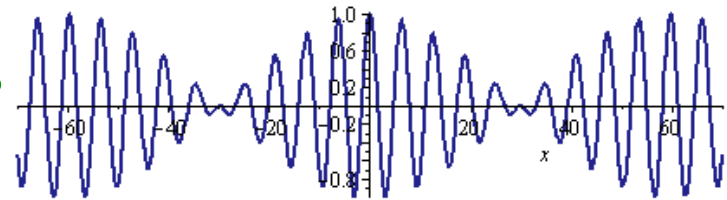
More Waves

- Two waves allow you to create localized “lumps”
- Three waves allow you to start separating these lumps
- More waves lets you get them farther and farther apart
- Infinity waves allows you to make the other lumps disappear to infinity – you have one lump, or a wave packet
- A single lump – a wave packet – looks and acts a lot like a particle

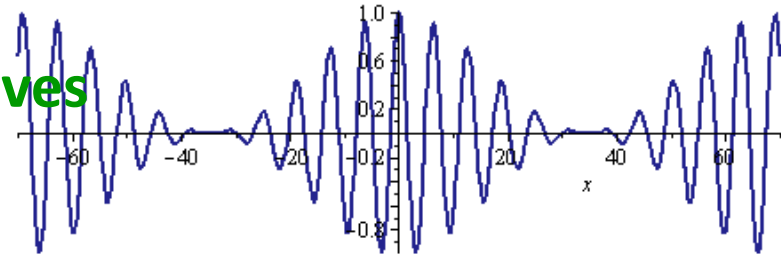
One wave



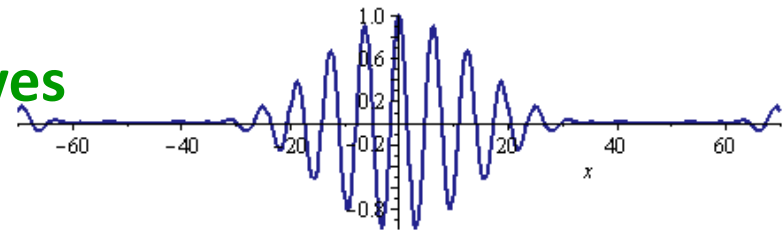
Two waves



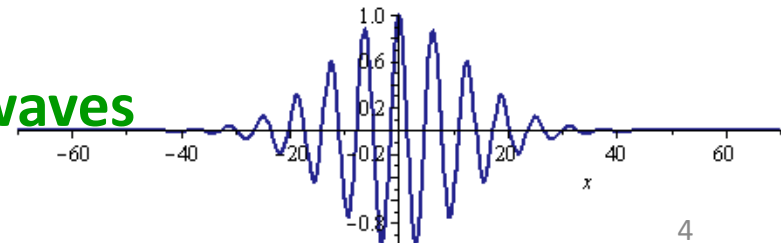
Three waves



Five waves

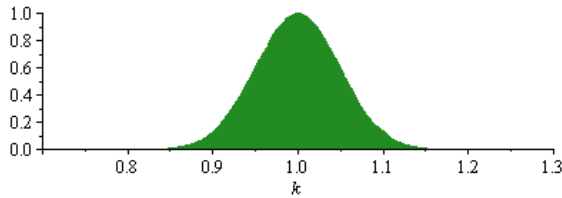


Infinity waves

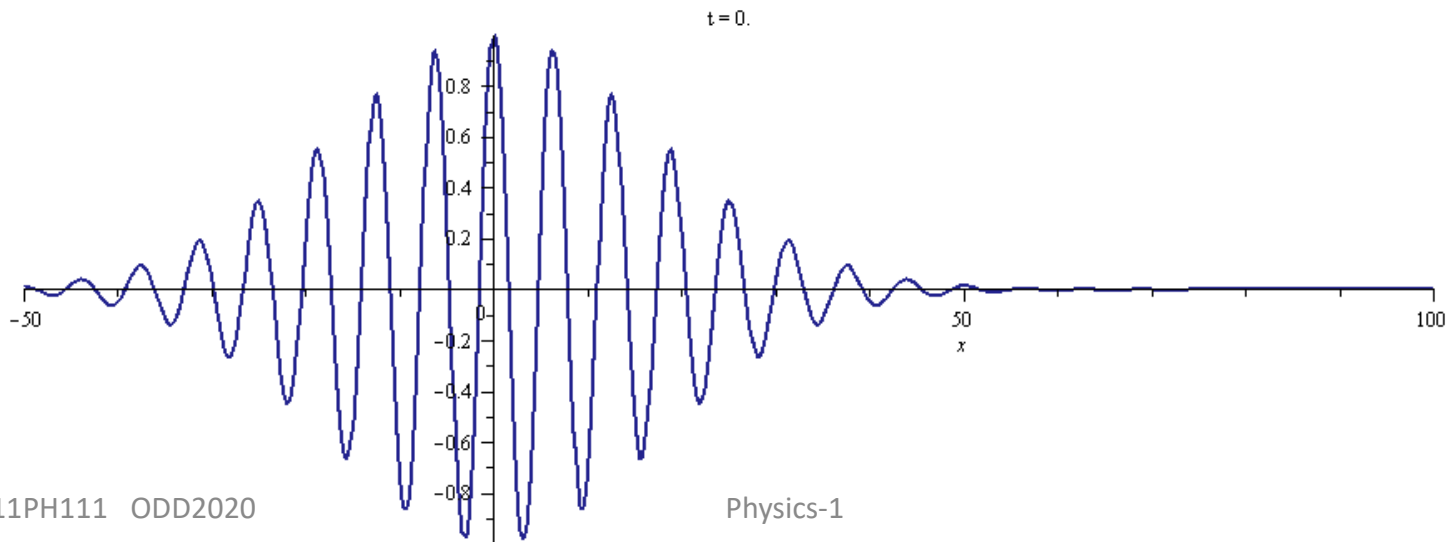


Wave Packets

- We can combine many waves to separate a “lump” from its neighbors
- With an infinite number of waves, we can make a wave packet
 - Contains continuum of wave numbers k
- Resulting wave travels and mostly stays together, like a particle
- Note both k -values and x -values have a spread Δk and Δx .



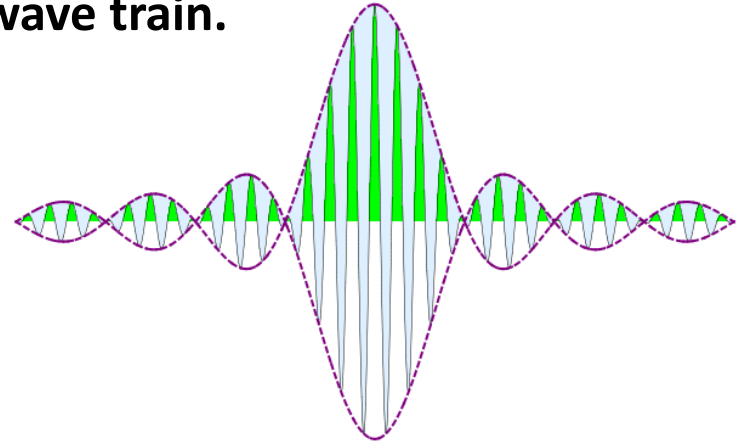
A wave packet is said to be **dispersive** if different frequencies travel at different speeds.



Wave (Phase) and Group Velocity

This discrepancy was overcome by considering that moving particle is associated with a '**wave packet**' rather than a simple wave train.

A wave packet is comprised of group of waves slightly differing in velocity and wavelengths with phase and amplitude such that they interfere constructively and destructively. **The velocity with which this wave moves forward in the medium is known as Group Velocity**



The average velocity of the individual monochromatic wave in the medium with which the wave packet is connected is known as Wave or Phase velocity.

If the velocities of the waves are the same, the velocity with which the wave group travels is the common phase velocity. However, if the phase velocity varies with wavelength, the different individual waves do not proceed together. This situation is called **dispersion**. As a result the wave group has a velocity different from the phase velocities of the waves that make it up.

Group Velocity

Compare to two wave formulas:

- Phase velocity formula is exactly the same, except we simply rename the average values of k and ω as simply k and ω
- Group velocity now involves very closely spaced values of k (and ω), and therefore we rewrite the differences as . . .

$$v_p = \frac{\bar{\omega}}{\bar{k}} \quad \rightarrow \quad v_p = \frac{\omega}{k}$$

$$v_g = \frac{\Delta\omega}{\Delta k} \quad \rightarrow \quad v_g = \frac{d\omega}{dk}$$

Theorem: Group velocity doesn't always equal phase velocity

$$\omega = kv_p$$

$$v_g = \frac{d\omega}{dk} = \frac{d(kv_p)}{dk} = v_p + \frac{dv_p}{dk} \neq v_p$$

Group Velocity

Wave nature: $\omega = 2\pi\nu = 2\pi(mc^2 / h) = 2\pi(m_0c^2 / h) / \sqrt{1 - v^2 / c^2}$

Particle nature: $k = 2\pi / \lambda = 2\pi(mv / h) = 2\pi(m_0v / h) / \sqrt{1 - v^2 / c^2}$

Group velocity is given by $v_g = \frac{d\omega}{dk} = \frac{d\omega / dv}{dk / dv}$

From above, we have $\frac{d\omega}{dv} = \frac{d}{dv} \left[\frac{2\pi m_0 c^2}{h \sqrt{1 - v^2 / c^2}} \right] = \frac{2\pi m_0 v}{h (1 - v^2 / c^2)^{3/2}}$

and $\frac{dk}{dv} = \frac{d}{dv} \left[\frac{2\pi m_0 v}{h \sqrt{1 - v^2 / c^2}} \right] = \frac{2\pi m_0}{h (1 - v^2 / c^2)^{3/2}}$

Hence, we can get group velocity $v_g = v$

Thus de-Broglie's wave group associated with a moving particle has the same velocity as the moving particle. The wave velocity (v_p) of the de-Broglie's wave evidently has no simple physical significance.

Relation between phase velocity and group velocity

As we know $v_g = \frac{d\omega}{dk}$ and $v_p = \frac{\omega}{k}$

$$v_g = \frac{d\omega}{dk} = \frac{d}{dk}(kv_p) \text{ \& } k = \frac{2\pi}{\lambda} \Rightarrow dk = -\frac{2\pi}{\lambda^2} d\lambda$$

$$v_g = \frac{d}{dk}(kv_p) = v_p + k \frac{dv_p}{dk} = v_p - \lambda \frac{dv_p}{d\lambda}$$

$$v_g = v_p - \lambda \frac{dv_p}{d\lambda}$$

$$v_g = v_p \Rightarrow \frac{dv_p}{dk} = 0$$

$$v_p = \frac{\omega}{k} = \frac{2\pi\nu}{2\pi/\lambda} = \nu\lambda = \frac{E}{h} \cdot \frac{h}{p} = \frac{mc^2}{mv} = \frac{c^2}{v}$$

since, group velocity is always equals to particle velocity ($v_g = v$), then $v_p v_g = c^2$

How to calculate them:

- You need the *dispersion relation*: the relationship between ω and k , with only constants in the formula
- Example: light in vacuum has $\omega = ck$

$$v_p = \frac{\omega}{k} = \frac{ck}{k} = c$$

$$v_g = \frac{d\omega}{dk} = \frac{d}{dk}(ck) = c$$

**If the dispersion relation is $\omega = Ak^2$,
with A a constant, what are the
phase and group velocity?**

$$v_p = \frac{\omega}{k} = \frac{Ak^2}{k} = Ak$$

$$v_g = \frac{d\omega}{dk} = \frac{d}{dk}(Ak^2) = 2Ak$$